

Turn Buckles

IS:3121-1981



EYE TO EYE



EYE TO HOOK



EYE TO JAW



JAW TO JAW



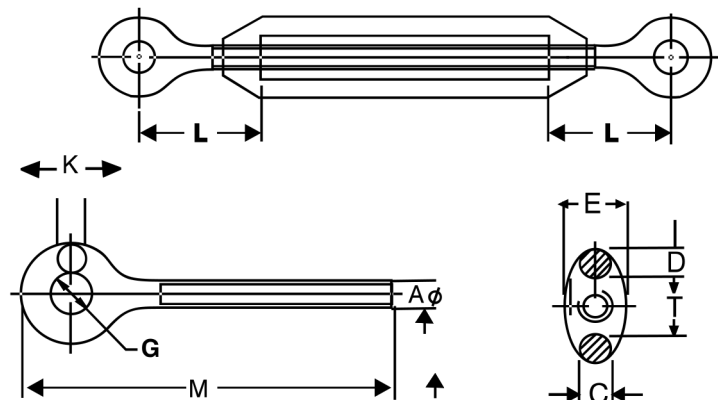
HOOK TO HOOK



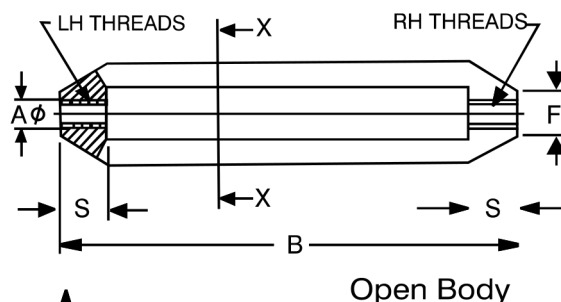
STUD TO STUD

Nominal Size		Assembly and Body					Screw Eye		Proof Load	Safe Working Load
A	B	D	E	L	S	G	K	M	kN	kN
M6	100	5	12	32	10	9	6	80	2.0	1.0
M8	125	6	16	35	13	13	8	100	3.0	1.5
M10	160	8	19	45	15	16	10	130	6.0	3.0
M12	200	10	22	54	18	19	12	150	9.0	4.0
M14	225	12	25	60	21	22	12	175	12.0	6.0
M16	250	14	29	66	24	24	14	200	15.2	7.6
M20	315	16	33	80	30	30	16	230	22.4	11.2
M22	355	18	41	88	32	33	18	270	32.4	16.2
M24	400	20	43	100	35	36	18	310	40.0	20.0
M30	450	22	50	120	42	44	24	350	63.0	31.5
M36	450	28	60	135	50	52	28	380	90.0	45.0
M45	450	36	75	165	60	67	35	415	142.0	71.0
M52	450	40	100	190	70	80	42	450	190.0	95.0

Nominal Size								
A Dia	J	K	L	M	N	P	Q	G
M12	12	9.5	30	165	115	29	12	M12
M16	19	9.5	50	185	115	32	12	M14
M20	22	12	60	195	115	35	16	M16
M24	25	12	65	240	150	42	19	M20
M27	29	12	70	270	175	48	22	M22
M30	32	14	75	280	175	54	25	M27
M33	35	14	85	300	190	60	25	M30
M36	39	16	90	315	190	60	29	M30
M39	42	19	100	335	200	66	32	M33
M45	52	22	115	350	200	80	38	M39
M52	54	25	125	370	200	95	45	M45
M56	60	29	125	375	200	100	48	M52
M60	70	29	135	380	200	110	52	M52
M64	60	29	135	508	290	127	52	M58
M68	67	28	145	553	310	142	56	M64
M75	73	30	160	596	330	157	64	M70
M80	79	32	175	640	350	175	72	M76
M90	84	34	190	684	370	193	76	M85



SECTION XX



Open Body

Specifications given are indicative and are subject to change.

Turn Buckles

Federal Specification FF-T-791B, Type 1



HOOK & HOOK

HOOK & EYE

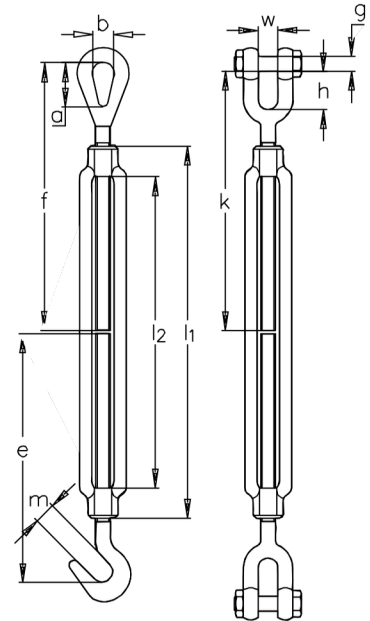
EYE & EYE

JAW & EYE

JAW & JAW

Nominal size *	I ₁ mm	I ₂ mm	m mm	e mm	a mm	b mm	f mm	w mm	h mm	g inch	k mm	Working Load Limit (Kg)		Weight Per Piece (Kg)		
												Eye & Eye	Hook & Hook	Eye & Eye	Jaw & Eye	Jaw & Jaw
												Eye & Jaw	Hook & Eye	Hook & Eye		
inch												Jaw & Jaw	Jaw & Jaw	Hook & Hook		
3/8X 6	180	152	12	129	28	13	137	12	22	1 5/16	137	540	450	0,39	0,42	0,45
1/2X 6	190	152	15	147	36	18	153	16	27	1 3/8	147	1000	680	0,67	0,72	0,76
1/2X 9	266	228	15	187	36	18	193	16	27	1 3/8	187	1000	680	0,84	0,89	0,93
1/2X 12	342	304	15	222	36	18	228	16	27	1 3/8	222	1000	680	1,01	1,06	1,10
5/8X 6	200	152	20	166	44	22	177	19	33	1 1/2	161	1590	1020	1,07	1,16	1,25
5/8X 9	276	228	20	206	44	22	217	19	33	1 1/2	201	1590	1020	1,31	1,40	1,49
5/8X 12	352	304	20	241	44	22	252	19	33	1 1/2	236	1590	1020	1,55	1,64	1,73
5/8X 18	505	457	-	-	-	-	-	19	33	1 1/2	316	1590	-	-	-	2,21
3/4X 6	210	152	23	181	54	25	196	23	38	1 5/8	173	2360	1360	1,75	1,89	2,03
3/4X 9	287	228	23	221	54	25	236	23	38	1 5/8	213	2360	1360	2,09	2,23	2,37
3/4X 12	362	304	23	256	54	25	271	23	38	1 5/8	248	2360	1360	2,42	2,56	2,70
3/4X 18	515	457	23	336	54	25	351	23	38	1 5/8	328	2360	1360	3,08	3,22	3,36
7/8X 12	372	304	26	273	60	31	287	28	44	1 3/4	266	3270	1810	3,56	3,80	4,04
7/8X 18	524	457	26	353	60	31	367	28	44	1 3/4	346	3270	1810	4,43	4,67	4,91
1 x 12	381	304	29	286	76	36	323	30	52	1 7/8	286	4540	2270	5,05	5,33	5,61
1 x 18	533	457	29	366	76	36	403	30	52	1 7/8	366	4540	2270	6,25	6,53	6,81
1 1/4X 12	387	304	-	-	90	46	360	44	71	1 1/8	330	6890	-	8,60	9,20	9,80
1 1/4X 18	540	457	-	-	90	46	440	44	71	1 1/8	380	6890	-	10,40	11,00	11,60
1 1/4X 24	693	610	-	-	90	46	495	44	71	1 1/8	479	6890	-	12,20	12,90	13,60
1 1/2X 12	400	304	-	-	104	54	390	52	71	1 3/8	360	9710	-	13,20	14,30	15,40
1 1/2X 18	550	457	-	-	104	54	465	52	71	1 3/8	430	9710	-	15,82	16,92	18,02
1 1/2X 24	703	610	-	-	104	54	540	52	71	1 3/8	496	9710	-	18,00	18,70	19,30
1 3/4X 18	570	457	-	-	-	-	-	60	86	1 3/4	440	12700	-	-	-	23,56
1 3/4X 24	720	610	-	-	117	59	577	60	86	1 3/4	500	12700	-	26,00	27,50	26,81
2 x 24	735	610	-	-	143	67	632	64	95	2 3/8	540	16780	-	37,58	40,80	45,04
2 1/2X 24	800	610	-	-	-	-	-	73	113	2 1/2	580	27220	-	-	-	72,32

* Proof load is 2 times Working Load Limit. Ultimate Load is 5 times Working Load Limit.
 Specifications given are indicative and are subject to change.

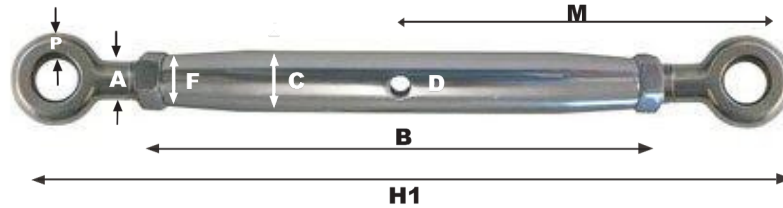


TURNBUCKLES RECOMMENDED
 FOR STRAIGHT OR IN-LINE PULL ONLY.

- Galvanized steel.
- Turnbuckle eyes are forged and elongated, by design, to maximize easy attachment in system and minimize stress in the eye. For turnbuckle size 1/4" through 2 1/2" a shackle one size smaller can be reeved through eye.
- Modified BSW thread for improved fatigue properties.
- Lock Nuts available for all sizes
- End fittings are Quenched and Tempered, bodies heat treated by normalizing

Rigging/ Bottle Screws

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Nominal Size A	Dia of Rope mm	H1		B	C Dia	D Dia	E	F Dia Min	K Dia	M	Proof Load kN	Safe Working Load kN
M12	8	330	525	230	27	10	4	20	12	177	10.0	5.0
M16	10	370	550	230	27	10	4	22	12	197	18.0	9.0
M20	12	400	570	230	34	10	4	25	16	211	28.0	14.0
M24	14	475	700	300	34	12	5	32	16	246	36.0	18.0
M27	16	550	825	360	43	12	5.5	35	20	290	44.0	22.0
M30	18	550	825	360	43	14	5.5	38	20	290	63.0	31.5
M33	20	600	875	380	48	14	6.5	44	25	325	75.0	37.5
M36	22	600	875	380	60	16	6.5	47	25	325	86.0	43.0
M39	25	660	960	400	60	16	6.5	50	25	350	100.0	50.0
M45	29	700	960	400	76	19	6.5	60	28	378	112.0	56.0
M52	32	750	1000	400	76	19	8	70	36	406	144.0	72.0
M56	35	775	1025	400	83	19	8	73	36	421	194.0	97.0
M60	38	800	1050	500	90	20	9.5	76	40	440	214.0	107.0
M64	41	1070	1450	580	95	20	13	91	47	580	286.0	143.0
M68	44	1120	1590	620	103	20	14	96	51	635	342.0	171.0
M75	48	1270	1700	660	111	20	14	103	57	690	400.0	200.0
M80	51	1360	1760	700	119	20	15	112	63	740	500.0	250.0
M90	54	1440	1860	740	127	20	16	122	69	790	624.0	312.0

Specifications given are indicative and are subject to change.

AVAILABLE IN THE FOLLOWING TYPES

JAW AND JAW TYPE



JAW AND EYE TYPE



EYE AND EYE TYPE



ALSO AVAILABLE IN EYE & HOOK / HOOK & HOOK TYPE